PSEUDO-COHERENCE DESCENDS ALONG A SURJECTION WITH NILPOTENT KERNEL

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Allowed standard prerequisites (editorial).

- A pseudo-coherent complex can be represented by a bounded above complex of finite free modules. invoked at proof lines 19505-19507. Produces the finite free complex E whose terms are the reductions of the complex to be built on the other side.
- A bounded above complex over the quotient with a lift of its terms can be lifted to a bounded above complex of projective modules with the prescribed reductions. invoked at proof lines 19507-19511. Supplies the complex P of projective modules lifting E, which is the candidate representing object on the other side.
- Nakayama's lemma lets one conclude that a projective module over the ring whose reduction to the quotient is finite free is itself finite free. invoked at proof lines 19512-19513. Upgrades the projectivity of the terms of P to finite freeness, which is what pseudo-coherence of the lifted object requires.

Source reference. Original lemma 0650: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16345–16345. Used at proof line 19504 for the direction that pseudo-coherence of the lifted object implies pseudo-coherence of K; the converse direction, which is the substance, is inside the counted span.

Source reference. Original lemma 064U: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16048–16048. Invoked at proof lines 19505-19507 to represent K by a finite free complex; the lifting and the Nakayama argument built on that representation are inside the counted span.

Source reference. Original lemma 09AR: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19469–19469. Used at proof lines 19507-19511 to lift the finite free complex to a projective one refining the given reduction; checking that the lifted complex has the right reductions and the final Nakayama step are inside the counted span.

Lemma 0.2. *Let $R' \to R$ be a surjective ring map whose kernel is a nilpotent ideal. Let $K' \in D(R')$ and set $K = K' \otimes^{\mathbf{L}}_{R'} R$. Then K is pseudo-coherent if and only if K' is pseudo-coherent.*

Proof. One direction follows from Lemma 0650. For the other direction, assume K is pseudo-coherent. Then by Lemma 064U we can represent K by a bounded above complex $E^\bullet$ of finite free R-modules. By Lemma 09AR we can represent K' by a

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bounded above complex $P^\bullet$ of projective R'-modules such that $P^n \otimes_{R'} R = E^n$. By Nakayama's lemma we see that P^n is finite free and we conclude that K' is pseudo-coherent as well. $\square$